

Hoang Minh Thanh

AI Agent Engineer

hmthanh.github.io [hmthanh](#) [in hmthanh](#) [hmthanh](#) [OCID](#) [Google scholar](#)

★ Keypoint

- **Strong Foundation in Generative AI and ML Development:** With hands-on experience developing transformer-based diffusion models for human motion synthesis in projects like [DeepGesture](#), I align well with responsibilities in developing core modules for generative AI agents and optimizing performance through techniques like prompt engineering and architecture design.
- **Proficiency in Rapid Prototyping and Full-Stack Development:** Skilled in front-end technologies [Million-Scope](#) (e.g., NextJS, Three.js, p5.js) for visualization and prototyping and fullstack building evaluation platforms
- **Experience Building Evaluation Systems and Backend Optimization:** One of organizer of the [GENEA Leaderboard](#), I designed a production-ready system for evaluating AI gesture models with user study workflows
- **Adaptability and Ownership in Dynamic Environments:** Proven through roles in experience Agile international teams showcasing a strong ownership mindset and ability to thrive in fast-paced settings while holding a MSc with research paper on arxiv and Pet project.

🏛 Education

AS	University of Science - VNUHCM, Information Technology	2012 – 2015
BS	University of Science - VNUHCM, Computer Science	Sept. 2018 – Sept. 2020
	• GPA: 3.3/4.0 • Thesis: GCAT - Link Prediction in Knowledge Graph, [Report], [Code]	
MSc	University of Science - VNUHCM, Computer Science	Oct. 2021 – Dec. 2024
	• GPA: 3.44/4.0 • Thesis: OpenHuman: A conversational gesture synthesis system based on emotions and semantics, [Report], [Code]	

📖 Publications

DeepGesture: A conversational gesture synthesis system based on emotions and semantics [arXiv] [Homepage]	Jul. 2025
Towards a GENE Leaderboard [arXiv]	Oct. 2024
Graph Collaborative Attention Network for Link Prediction in Knowledge Graphs [arXiv] [Homepage]	Oct. 2020

🛠 Experience

Rakumo Inc, Software Engineer	Jul. 2019 – Apr. 2020
• Developed a Django-based web application with integrated DocuSign.	
FPT Software, AWS Data Engineer	Oct. 2020 – Nov. 2021
• CI/CD data pipeline: implemented and modified real-time and batch data pipelines on AWS using Kinesis, Lambda, S3, Glue, StepFunction, SNS, ..etc.. • Cox Automotive (onsite-offshore): Collaborated with international teams in Agile	

VNG Corp, Software Engineer

Oct. 2021 – May 2024

ZDN Team (Zalo Content Delivery Network): caching service serve millions request

- Implement scheduled `ThreadPool` for monitoring stats of server execution.
- Built real-time server status with visualization via `amchart`, `p5.js` in `NextJS`.
- Integrate outlier detection to prevent DDoS attacks in the caching server.

OpenHuman, Creator

May 2024 – Present

- Join GENE Research Team ([GENEA Leaderboard](#), [HEMVIP](#)): Designed and built a full-stack, production-ready leaderboard system to evaluate multiple AI gesture generation models, integrating user study workflows via `Prolific`.

</> Pet Projects

MillionScope ([Code](#)) (Fullstack self-made)

[millionscope.com](#) [↗](#)

- Custom Reasoning LLM Chat with ChatGPT-Style Interface, Streaming Responses, and Result Annotations.

SurveyWithCode ([Code](#)) (In progress)

[surveywithcode.dev](#) [↗](#)

- Remake `PapersWithCode` using `Nextjs` and `d3.js`

Three.js realisitic face ([Code](#)): Loaded and scaled facial OBJ parts in Blender, applied edited texture maps (albedo, roughness, specular, clearcoat) and tone mapping

[renderengine.pages.dev](#) [↗](#)

Unity Game Pucca Runner ([Code](#)) Pathfinding with a 2D hash map and Dijkstra's algorithm; runner moves toward the player and triggers collision on boundary breach

[Pucca Runner](#) [↗](#)

Others: [Morph target animation](#), [MetaPet](#) ([metapet.vercel.app](#)), [cortexpod.com](#), [pithagon.com](#)

🌟 Certifications

- [Critical Thinking Certification 2014](#) [↗](#), [Emotion Recognition Certification 2019](#) [↗](#), [Summer ML](#) [↗](#)
- Coursera: [Design Pattern](#) [↗](#), [TensorFlow Developer Professional](#) [↗](#) 2021
- **AWS:** [Solutions Architect - Associate](#) [↗](#), [Machine Learning - Specialty](#) [↗](#); [Solution Architect](#) [↗](#) 2021
- [Solution Architect](#) [↗](#) 2021

💻 Technologies

- **AI/ML & Agent Development:** Python, PyTorch, LangChain, Prompt Engineering, RAG frameworks, Agent system design patterns, MCP protocol
- **Frontend:** React, Next.js, TypeScript, Tailwind CSS v4, D3.js, WebSockets, RESTful APIs
- **Backend:** Node.js, PostgreSQL, Redis, MongoDB, Docker, AWS (Lambda, EC2, S3), message queues, microservices architecture
- **Evaluation & Monitoring:** MLflow, Weights & Biases, CI/CD (GitHub Actions, Jenkins)
- **Tools:** Git, VS Code, Jupyter, Postman, Agile/Scrum methodologies